Create By Saurav Bhosale

Sql Project On Pizza Sales :-

```
CREATE DATABASE pizza_project;

USE pizza_project;

-- ========================

-- Table: pizza_type

-- ========================

CREATE TABLE pizza_type (

   pizza_type_id VARCHAR(50) PRIMARY KEY,

   name TEXT NOT NULL,

   category TEXT NOT NULL,

   ingredients TEXT NOT NULL

);
```

pizza_type_id	name	category	ingredients
bbq_ckn	The Barbecue Chicken Pizza	Chicken	Barbecued Chicken, Red Peppers, Green Peppe...
cali_ckn	The California Chicken Pizza	Chicken	Chicken, Artichoke, Spinach, Garlic, Jalapeno P...
ckn_alfredo	The Chicken Alfredo Pizza	Chicken	Chicken, Red Onions, Red Peppers, Mushrooms...
ckn_pesto	The Chicken Pesto Pizza	Chicken	Chicken, Tomatoes, Red Peppers, Spinach, Garl...
southw_ckn	The Southwest Chicken Pizza	Chicken	Chicken, Tomatoes, Red Peppers, Red Onions, ...
thai_ckn	The Thai Chicken Pizza	Chicken	Chicken, Pineapple, Tomatoes, Red Peppers, T...
big_meat	The Big Meat Pizza	Classic	Bacon, Pepperoni, Italian Sausage, Chorizo Sau...
classic_dlx	The Classic Deluxe Pizza	Classic	Pepperoni, Mushrooms, Red Onions, Red Peppe...
hawaiian	The Hawaiian Pizza	Classic	Sliced Ham, Pineapple, Mozzarella Cheese
ital_cpcllo	The Italian Capocollo Pizza	Classic	Capocollo, Red Peppers, Tomatoes, Goat Chee...
napolitana	The Napolitana Pizza	Classic	Tomatoes, Anchovies, Green Olives, Red Onion...
pep_msh_pep	The Pepperoni, Mushroom, ...	Classic	Pepperoni, Mushrooms, Green Peppers
pepperoni	The Pepperoni Pizza	Classic	Mozzarella Cheese, Pepperoni
the_greek	The Greek Pizza	Classic	Kalamata Olives, Feta Cheese, Tomatoes, Garli...
brie_carre	The Brie Carre Pizza	Supreme	Brie Carre Cheese, Prosciutto, Caramelized Oni...
calabrese	The Calabrese Pizza	Supreme	'Nduja Salami, Pancetta, Tomatoes, Red Onions...

pizza_types 1 ×

```
-- Table: pizzas

-- ========================

CREATE TABLE pizzas (

   pizza_id VARCHAR(50) PRIMARY KEY,

   pizza_type_id VARCHAR(50) NOT NULL,

   size VARCHAR(5),

   price DECIMAL(5,2),


   FOREIGN KEY (pizza_type_id)

   REFERENCES pizza_type(pizza_type_id));
```

Result Grid | Filter Rows:

pizza_id	pizza_type_id	size	price
bbq_ckn_s	bbq_ckn	S	12.75
bbq_ckn_m	bbq_ckn	M	16.75
bbq_ckn_l	bbq_ckn	L	20.75
cali_ckn_s	cali_ckn	S	12.75
cali_ckn_m	cali_ckn	M	16.75
cali_ckn_l	cali_ckn	L	20.75
ckn_alfredo_s	ckn_alfredo	S	12.75
ckn_alfredo_m	ckn_alfredo	M	16.75
ckn_alfredo_l	ckn_alfredo	L	20.75
ckn_pesto_s	ckn_pesto	S	12.75
ckn_pesto_m	ckn_pesto	M	16.75
ckn_pesto_l	ckn_pesto	L	20.75
southw_ckn_s	southw_ckn	S	12.75
southw_ckn_m	southw_ckn	M	16.75
southw_ckn_l	southw_ckn	L	20.75
thai_ckn_s	thai_ckn	S	12.75

pizzas 1 ×

-- Table: orders

-- ========================

CREATE TABLE orders (

order_id INT PRIMARY KEY,

order_date DATE NOT NULL,

order_time TIME NOT NULL

);

Result Grid | Filter Rows:

order_id	order_date	order_time
1	2015-01-01	11:38:36
2	2015-01-01	11:57:40
3	2015-01-01	12:12:28
4	2015-01-01	12:16:31
5	2015-01-01	12:21:30
6	2015-01-01	12:29:36
7	2015-01-01	12:50:37
8	2015-01-01	12:51:37
9	2015-01-01	12:52:01
10	2015-01-01	13:00:15
11	2015-01-01	13:02:59
12	2015-01-01	13:04:41
13	2015-01-01	13:11:55
14	2015-01-01	13:14:19
15	2015-01-01	13:33:00
16	2015-01-01	13:34:07

orders 1 ×

```
-- Table: order_details
-- ========================
CREATE TABLE order_details (
  order_detail_id INT PRIMARY KEY,
  order_id INT NOT NULL,
  pizza_id VARCHAR(50) NOT NULL,
  quantity INT NOT NULL,
  FOREIGN KEY (order_id)
  REFERENCES orders(order_id),
  FOREIGN KEY (pizza_id)
  REFERENCES pizzas(pizza_id)
);
```

Result Grid | Filter Rows: | Edit

order_detail_id	order_id	pizza_id	quantity
1	1	hawaiian_m	1
2	2	classic_dlx_m	1
3	2	five_cheese_l	1
4	2	ital_supr_l	1
5	2	mexicana_m	1
6	2	thai_ckn_l	1
7	3	ital_supr_m	1
8	3	prsc_argla_l	1
9	4	ital_supr_m	1
10	5	ital_supr_m	1
11	6	bbq_ckn_s	1
12	6	the_greek_s	1
13	7	spinach_supr_s	1
14	8	spinach_supr_s	1
15	9	classic_dlx_s	1
16	9	green_garde...	1

order_details 1 ×

Basic:

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1) Retrieve the total number of orders placed.

```
SELECT
  COUNT(order_id) AS Total_Orders
FROM
  pizza_project.orders;
```

Result Grid

Total_Orders
21350

2) Calculate the total revenue generated from pizza sales.

```
SELECT
   ROUND(SUM(order_details.quantity * pizzas.price),
         2) AS Total_Sales
FROM
   order_details
     JOIN
   pizzas ON pizzas.pizza_id = order_details.pizza_id;
```

Result Grid

Total_Sales
817860.05

3) Identify the highest-priced pizza.

```
SELECT
   pizza_types.name, pizzas.price AS Highest_Price
FROM
   pizza_types
     JOIN
   pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
ORDER BY Highest_Price DESC
LIMIT 1;
```

Result Grid | Filter Rows:

name	Highest_Price
The Greek Pizza	35.95

4) Identify the most common pizza size ordered.

```
SELECT
   pizzas.size,
   COUNT(order_details.order_detail_id) AS Order_Count
FROM
   pizzas
     JOIN
   order_details ON pizzas.pizza_id = order_details.pizza_id
GROUP BY pizzas.size
ORDER BY Order_Count DESC;
```

Result Grid | Filter

size	Order_Count
L	18526
M	15385
S	14137
XL	544
XXL	28

5) List the top 5 most ordered pizza types along with their quantities.

```
SELECT
    pizza_types.name, SUM(order_details.quantity) AS Quantity
FROM
    pizza_types
        JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
        JOIN
    order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.name
ORDER BY Quantity DESC
LIMIT 5;
```

Result Grid | Filter Rows:

name	Quantity
The Classic Deluxe Pizza	2453
The Barbecue Chicken Pizza	2432
The Hawaiian Pizza	2422
The Pepperoni Pizza	2418
The Thai Chicken Pizza	2371

Intermediate:

6) Join the necessary tables to find the total quantity of each pizza category ordered.

```
SELECT
    pizza_types.category AS category,
    SUM(order_details.quantity) AS quantity
FROM
    pizza_types
        JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
        JOIN
    order_details ON pizzas.pizza_id = order_details.pizza_id
GROUP BY category;
```

Result Grid | Filt

category	quantity
Classic	14888
Veggie	11649
Supreme	11987
Chicken	11050

7) Determine the distribution of orders by hour of the day.

```
SELECT
    HOUR(order_time), COUNT(orders.order_id)
FROM
```

```
    orders
GROUP BY HOUR(order_time);
```

Result Grid | Filter Rows:

hour(order_time)	count(orders.order_id)
11	1231
12	2520
13	2455
14	1472
15	1468
16	1920
17	2336
18	2399
19	2009
20	1642
21	1198
22	663
23	28
10	8

8) Join relevant tables to find the category-wise distribution of pizzas.

```
SELECT
    category, COUNT(name)
FROM
    pizza_types
GROUP BY category;
```

Result Grid | Filter Rows:

category	COUNT(name)
Chicken	6
Classic	8
Supreme	9
Veggie	9

9) Group the orders by date and calculate the average number of pizzas ordered per day.

```
SELECT
    ROUND(AVG(quantity), 0)
FROM
    (SELECT
        orders.order_date, SUM(order_details.quantity) AS quantity
    FROM
        orders
    JOIN order_details ON orders.order_id = order_details.order_id
    GROUP BY orders.order_date) AS order_quantity;
```

Result Grid | Filter Rows:

ROUND(AVG(quantity), 0)
138

10) Determine the top 3 most ordered pizza types based on revenue.

```
SELECT
    pizza_types.name,
    SUM(order_details.quantity * pizzas.price) AS revenue
FROM
```

```
    pizza_types
        JOIN
    pizzas ON pizzas.pizza_type_id = pizza_types.pizza_type_id
        JOIN
    order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.name
ORDER BY revenue DESC
LIMIT 3;
```

Result Grid | Filter Rows:

name	revenue
The Thai Chicken Pizza	43434.25
The Barbecue Chicken Pizza	42768
The California Chicken Pizza	41409.5

Advanced:

11) Calculate the percentage contribution of each pizza type to total revenue.

```
SELECT
    pizza_types.category,
    ROUND(SUM(order_details.quantity * pizzas.price) / (SELECT
                ROUND(SUM(order_details.quantity * pizzas.price),
                        2) AS Total_Sales
            FROM
                order_details
                    JOIN
                pizzas ON pizzas.pizza_id = order_details.pizza_id) * 100,
            2) AS revenue
FROM
    pizza_types
        JOIN
    pizzas ON pizzas.pizza_type_id = pizza_types.pizza_type_id
        JOIN
    order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.category
ORDER BY revenue DESC;
```

Result Grid | Filter Rows:

category	revenue
Classic	26.91
Supreme	25.46
Chicken	23.96
Veggie	23.68

12) Analyze the cumulative revenue generated over time.

```
select order_date,
sum(revenue) over (order by order_date) as cum_revenue
```

```
from
( select orders.order_date,
sum( order_details.quantity * pizzas.price) as revenue
from order_details join pizzas on order_details.pizza_id = pizzas.pizza_id
join orders
 on orders.order_id = order_details.order_id
 group by orders.order_date) as sales;
```

Result Grid | Filter Rows:

order_date	cum_revenue
2015-01-05	11929.55
2015-01-06	14358.5
2015-01-07	16560.7
2015-01-08	19399.05
2015-01-09	21526.4
2015-01-10	23990.350000000002
2015-01-11	25862.65
2015-01-12	27781.7
2015-01-13	29831.300000000003
2015-01-14	32358.700000000004
2015-01-15	34343.50000000001
2015-01-16	36937.65000000001
2015-01-17	39001.75000000001
2015-01-18	40978.600000000006
2015-01-19	43365.75000000001
2015-01-20	45763.65000000001

13) Determine the top 3 most ordered pizza types based on revenue for each pizza category

```
select Name, revenue from(

 select Category, Name , revenue,

rank () over (partition by Category order by revenue desc ) as rn

from(

SELECT

   pizza_types.category AS Category,

   pizza_types.name as Name,

   SUM(order_details.quantity * pizzas.price) AS revenue

FROM

   pizza_types

     JOIN

   pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id

     JOIN
```

```
    order_details ON pizzas.pizza_id = order_details.pizza_id
GROUP BY Category, Name ) as a ) as b
where rn <=3;
```

Name	revenue
The Thai Chicken Pizza	43434.25
The Barbecue Chicken Pizza	42768
The California Chicken Pizza	41409.5
The Classic Deluxe Pizza	38180.5
The Hawaiian Pizza	32273.25
The Pepperoni Pizza	30161.75
The Spicy Italian Pizza	34831.25
The Italian Supreme Pizza	33476.75
The Sicilian Pizza	30940.5
The Four Cheese Pizza	32265.70000000065
The Mexicana Pizza	26780.75
The Five Cheese Pizza	26066.5